

Process attribute charts

Process classes and class members

A *process* is a method of shaping, finishing or joining a material. *Sand casting*, *injection molding*, *fusion welding* and *polishing* are all processes. The choice, for a given component, depends on the material of which it is to be made, on its size, shape and precision, and on how many are required

The manufacturing processes of engineering fall into nine broad classes:

Process classes	
Casting	(sand, gravity, pressure, die, etc)
Pressure molding	(direct, transfer, injection, etc)
Deformation processes	(rolling, forging, drawing, etc)
Powder methods	(slip cast, sinter, hot press, hip)
Special methods	(CVD, electroform, lay up, etc)
Machining	(cut, turn, drill, mill, grind, etc)
Heat treatment	(quench, temper, solution treat, age, etc)
Joining	(bolt, rivet, weld, braze, adhesives)
Surface finish	(polish, plate, anodise, paint)

Each process is characterised by a set of *attributes*: the materials it can handle, the shapes it can make and their precision, complexity and size and so forth. Process Selection Charts map the attributes, showing the ranges of size, shape, material, precision and surface finish of which each class of process is capable. They are used in the way described in "Materials Selection in Mechanical Design". The procedure does not lead to a final choice of process. Instead, it identifies a subset of processes which have the potential to meet the design requirements. More specialised sources must then be consulted to determine which of these is the most economical.

The hard-copy versions, shown here, are necessarily simplified, showing only a limited number of processes and attributes. Computer implementation, as in the CES Edu software, allows exploration of a much larger number of both.

Chart P1 The Process – Material matrix.

A given process can shape, or join, or finish some materials but not others. The matrix shows the links between material and process classes. A red dot indicates that the pair are compatible. Processes that cannot shape the material of choice are non-starters. The upper section of the matrix describes shaping processes. The two sections at the bottom cover joining and finishing.

		Metals, ferrous	Metals, non-ferrous	Ceramics	Glasses	Elastomers	Thermoplastics	Thermosets	Polymer foams	Composites
Shaping	Sand casting	•	•							
	Die casting	•	•							
	Investment casting	•	•							
	Low pressure casting	•	•							
	Forging	•	•							
	Extrusion		•							
	Sheet forming	•	•							
	Powder methods	•	•	•						
	Electro-machining	•	•	•						
	Conventional machining	•	•		•	•	•	•	•	
	Injection molding				•	•	•	•	•	
	Blow molding				•		•			
	Compression molding				•	•	•	•		
	Rotational molding					•	•	•	•	
	Thermo-forming					•	•	•		
	Polymer casting					•	•	•	•	
	Resin-transfer molding						•	•	•	•
	Filament winding									•
	Lay-up methods									•
	Vacuum bag									•
Joining	Adhesives	•	•	•	•	•	•	•	•	•
	Welding, metals	•	•							
	Welding, polymers					•	•	•	•	
	Fasteners	•	•	•	•	•	•	•	•	•
Finishing	Precision machining	•	•				•	•		•
	Grinding	•	•	•	•					•
	Lapping	•	•	•	•					•
	Polishing	•	•	•	•		•	•		•

Chart P2 The Process – Shape matrix.

Shape is the most difficult attribute to characterize. Many processes involve rotation or translation of a tool or of the workpiece, directing our thinking towards axial symmetry, translational symmetry, uniformity of section and such like. *Turning* creates axisymmetric (or circular) shapes; *extrusion*, *drawing* and *rolling* make prismatic shapes, both circular and non-circular. *Sheet-forming* processes make flat shapes (stamping) or dished shapes (drawing). Certain processes can make 3-dimensional shapes, and among these some can make hollow shapes whereas others cannot.

The process-shape matrix displays the links between the two. If the process cannot make the desired shape, it may be possible to combine it with a secondary process to give a process-chain that adds the additional features: casting followed by machining is an obvious example.

Information about material compatibility is included at the extreme left.

		Circular prismatic	Non-circular prismatic	Flat sheet	Dished sheet	3-D solid	3-D hollow
Ceramic shaping Polymer shaping Composite shaping	Metal shaping	Sand casting	●	●		●	●
		Die casting	●	●		●	●
		Investment casting	●	●		●	●
		Low pressure casting	●	●		●	●
		Forging	●	●		●	
		Extrusion	●	●			
	Ceramic shaping	Sheet forming	●	●	●		
		Powder methods	●	●		●	●
		Electro-machining	●	●	●	●	●
		Conventional machining	●	●	●	●	●
	Polymer shaping	Injection molding	●	●		●	●
		Blow molding			●		●
		Compression molding		●	●		
		Rotational molding			●		●
	Composite shaping	Thermo-forming			●		
		Polymer casting	●	●		●	●
		Resin-transfer molding	●	●	●	●	●
		Filament winding	●	●	●		●
		Lay-up methods		●	●	●	
		Vacuum bag		●	●		

Chart P3 The Process – Mass-range chart.

The bar-chart shows the typical mass-range of components that each processes can make. It is one of four, allowing application of constraints on size (measured by mass), section thickness, tolerance and surface roughness. Large components can be built up by joining smaller ones. For this reason the ranges associated with joining are shown in the lower part of the figure. In applying a constraint on mass, we seek single shaping-processes or shaping-joining combinations capable of making it, rejecting those that cannot.

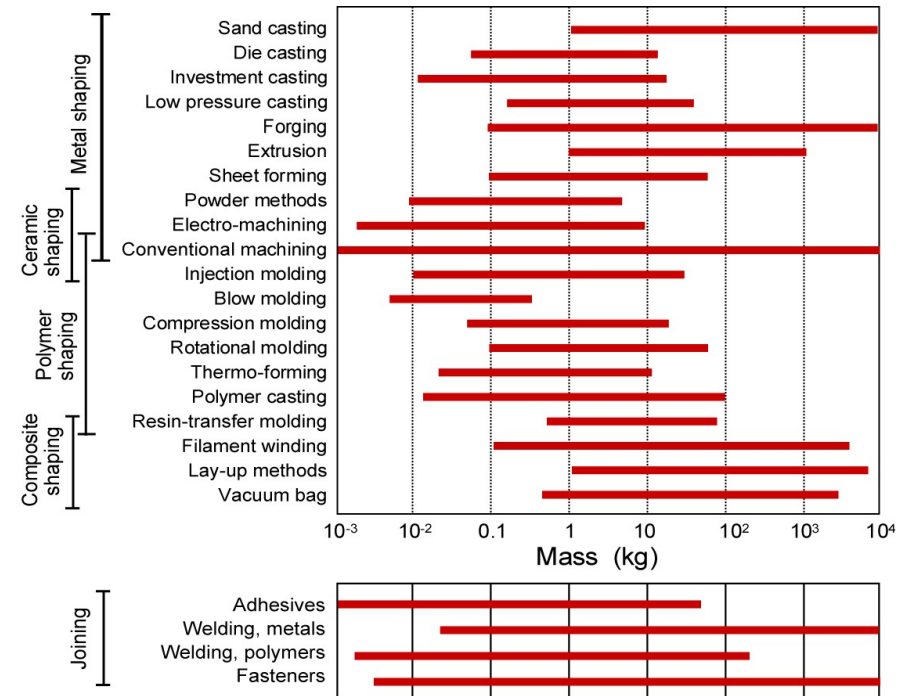


Chart P4 The Process – Section thickness chart.

The bar-chart on the right allows selection to meet constraints on section thickness. Surface tension and heat-flow limit the minimum section of gravity cast shapes. The range can be extended by applying a pressure or by pre-heating the mold, but there remain definite lower limits for the section thickness. Limits on rolling and forging-pressures set a lower limit on thickness achievable by deformation processing. Powder-forming methods are more limited in the section thicknesses they can create, but they can be used for ceramics and very hard metals that cannot be shaped in other ways. The section thicknesses obtained by polymer-forming methods – injection molding, pressing, blow-molding, etc – depend on the viscosity of the polymer; fillers increase viscosity, further limiting the thinness of sections. Special techniques, which include electro-forming, plasma-spraying and various vapour – deposition methods, allow very slender shapes.

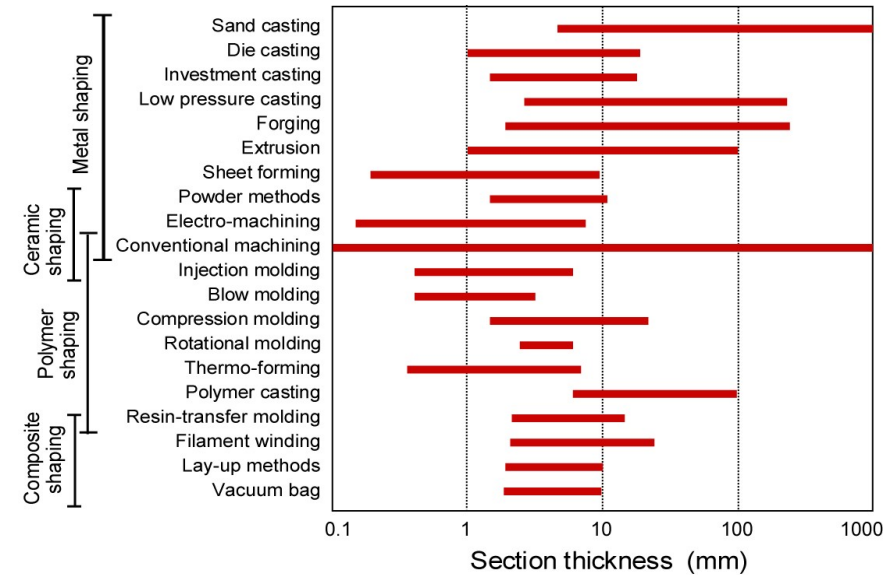


Chart P5 The Process – Tolerance chart.

No process can shape a part *exactly* to a specified dimension. Some deviation Δx from a desired dimension x is permitted; it is referred to as the *tolerance*, T , and is specified as

$x = 100 \pm 0.1$ mm, or as $x = 50^{+0.01}_{-0.001}$ mm. This bar chart allows selection to achieve a given tolerance.

The inclusion of finishing processes allows simple process chains to be explored

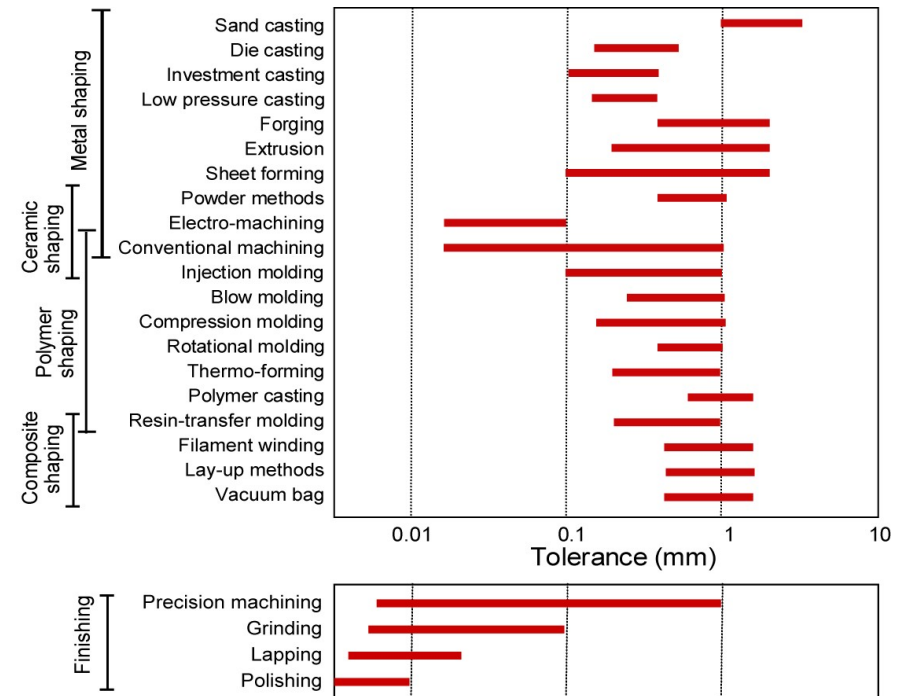


Chart P6 The Process – Surface roughness chart.

The *surface roughness* R , is measured by the root-mean-square amplitude of the irregularities on the surface. It is specified as $R < 100 \mu\text{m}$ (the rough surface of a sand casting) or $R < 0.01 \mu\text{m}$ (a highly polished surface). The bar chart on the right allows selection to achieve a given surface roughness.

The inclusion of finishing processes allows simple process chains to be explored.

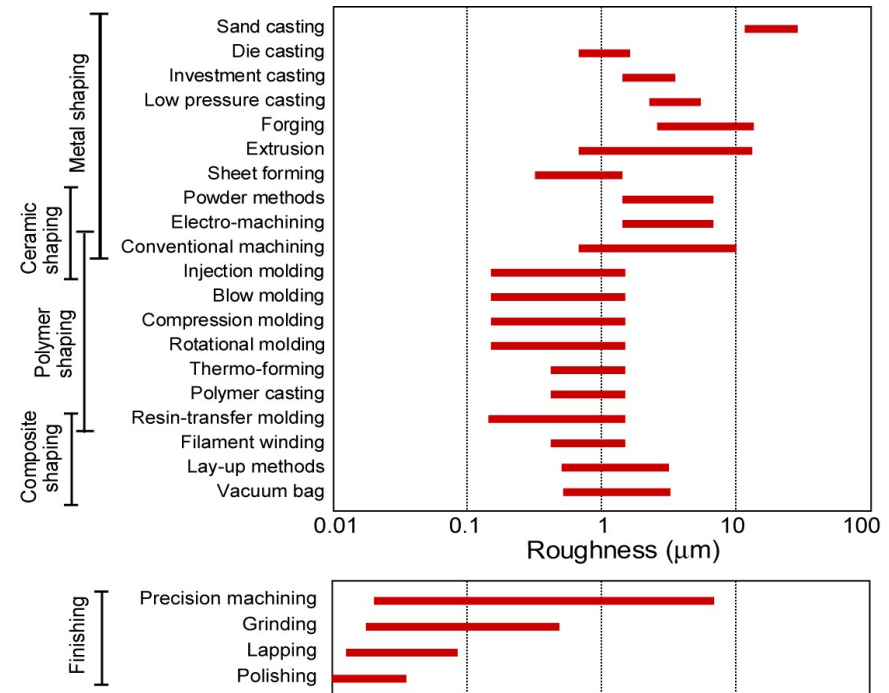


Chart P7 The Process – Economic batch-size chart.

Process cost depends on a large number of independent variables. The influence of many of the inputs to the cost of a process are captured by a single attribute: the *economic batch size*. A process with an economic batch size with the range $B_1 - B_2$ is one that is found by experience to be competitive in cost when the output lies in that range.

